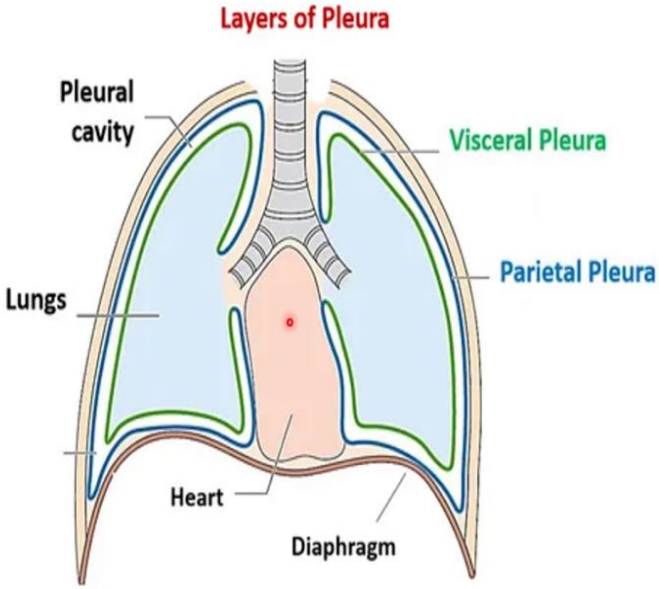
Gross Anatomy of Pleura and Lung



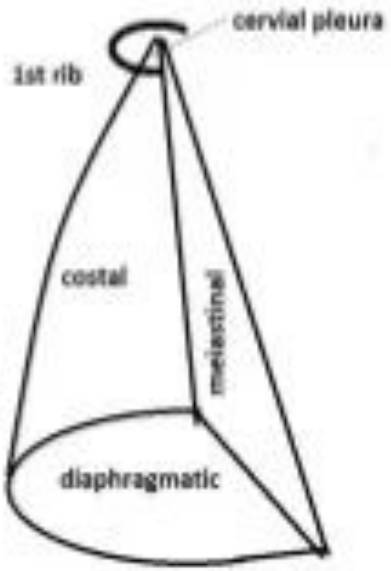
PLEURA:

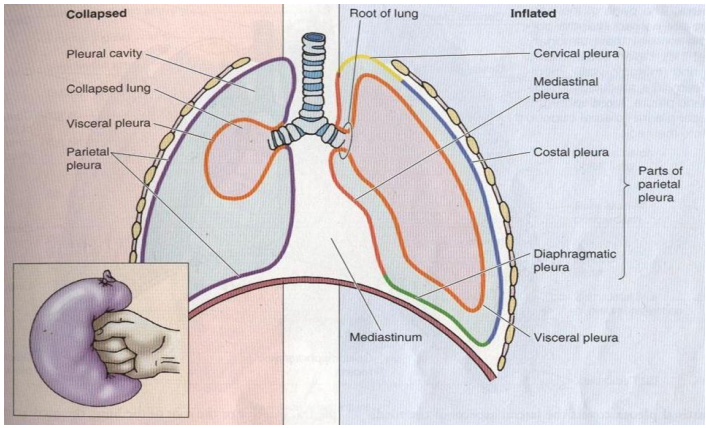
- Pleura is a **serous membrane lined by mesothelium** (simple squamous epithelium).
- There are **two pleural sacs on either side of mediastinum** i.e. right and left.
- Each sac is invaginated by the lung from its medial side.
- Thus, the **pleura has two layers: an outer/parietal layer and an inner/visceral layer.**
- The two layers are continuous with each other at hilum of the lung.
- Potential space between parietal and visceral layers is called pleural cavity.

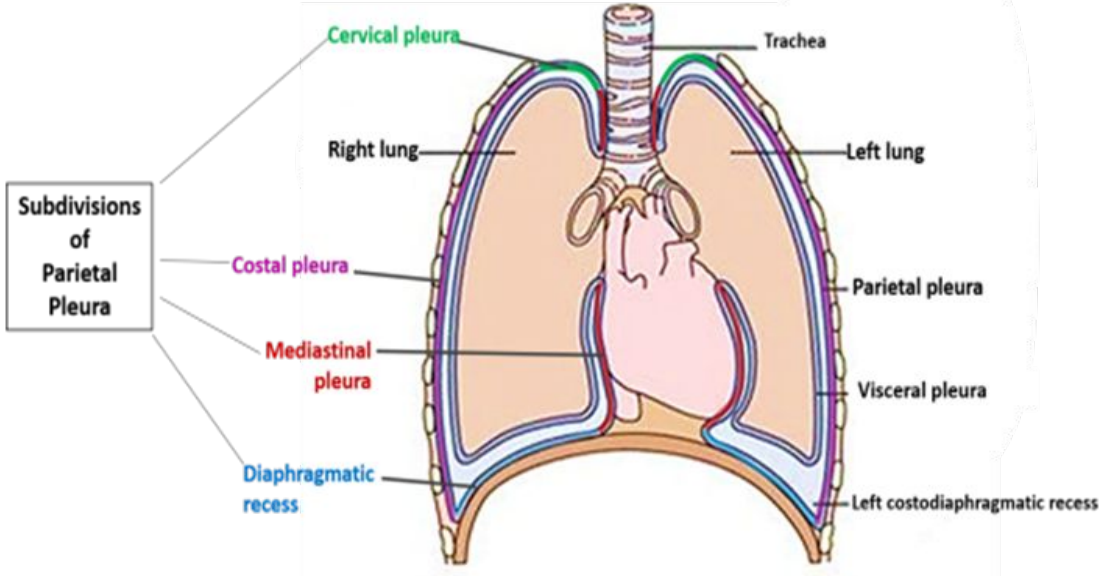


Parietal Pleura

1. **Mediastinal pleura** – Covers the lateral aspect of the mediastinum (the central component of the thoracic cavity, containing a number of organ).
2. **Cervical pleura** – Lines the extension of the pleural cavity into the neck.
3. **Costal pleura** – Covers the inner aspect of the ribs, costal cartilages, and intercostal muscles.
4. **Diaphragmatic pleura** – Covers the thoracic (superior) surface of the diaphragm.

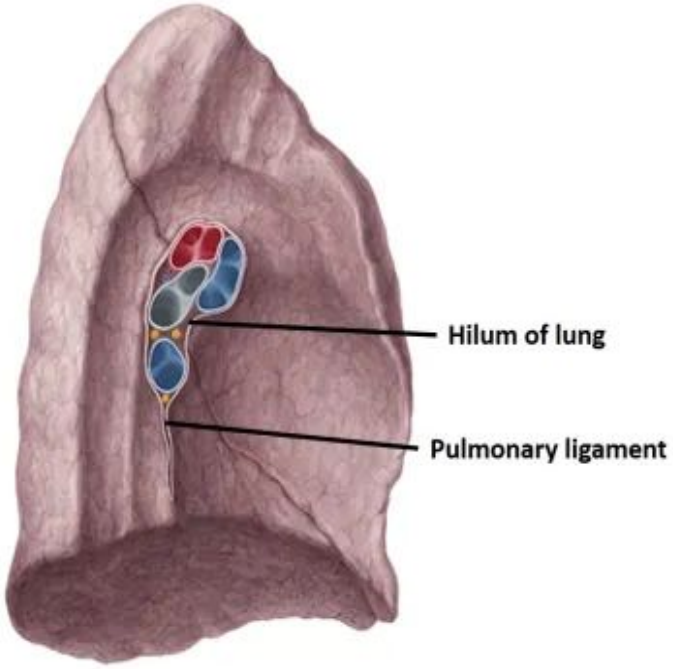


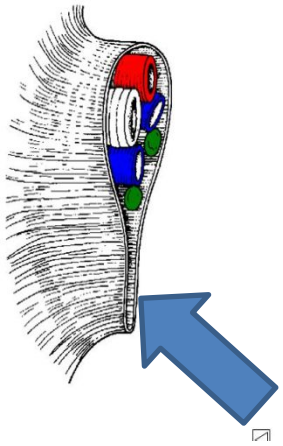




Pulmonary Ligament:

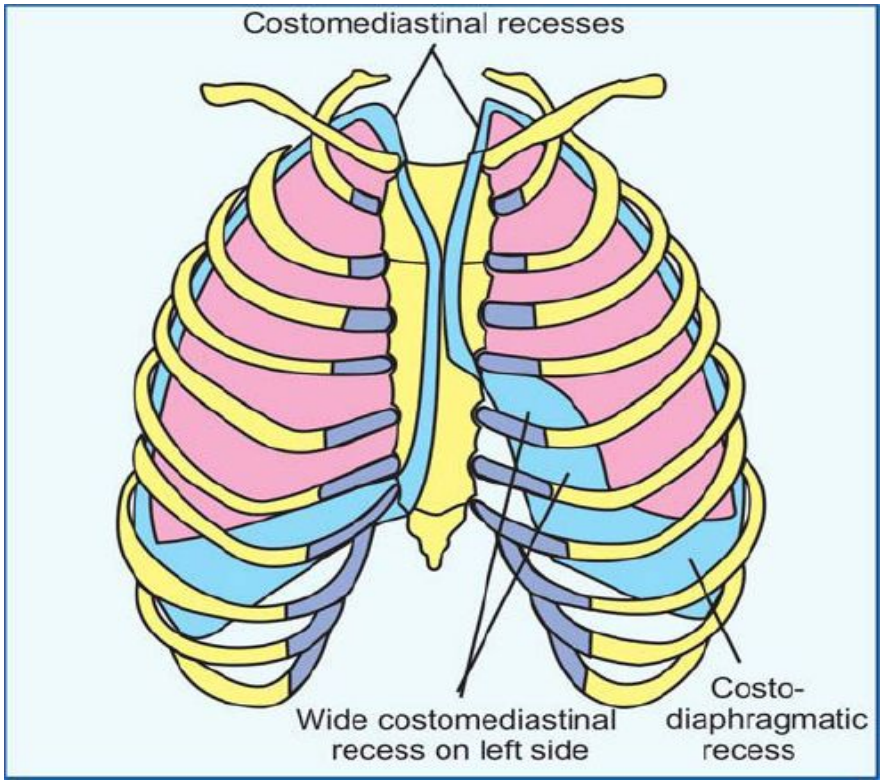
- The parietal pleura surrounding the root of the lung extends downwards beyond the root as a fold called pulmonary ligament.
- It contains areolar tissue and a few lymphatics.
- It serves as a dead-space into which pulmonary veins can expand during increased venous return.





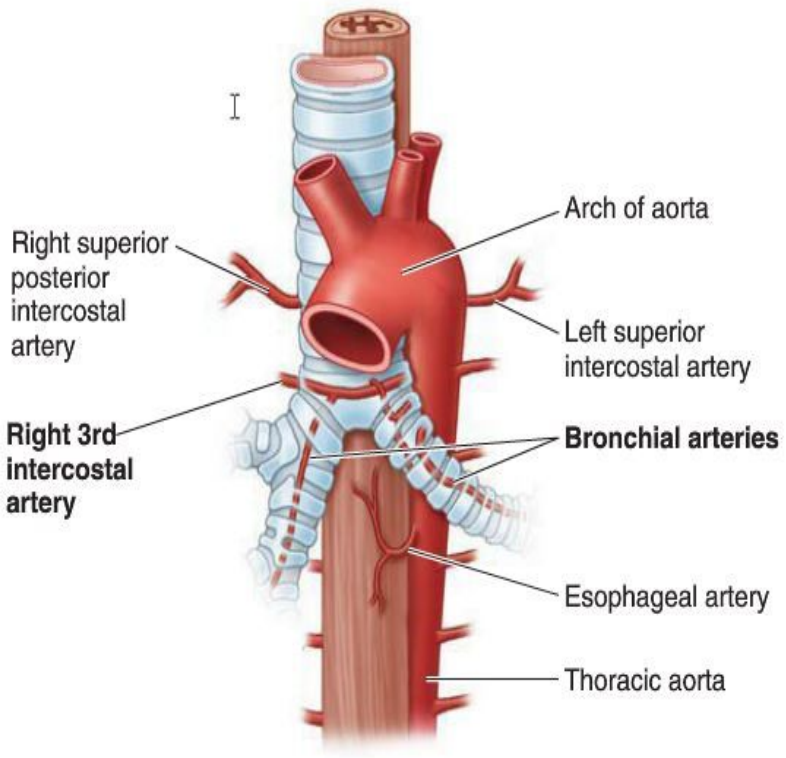
Recesses of Pleura:

- These are the **reserve spaces in parietal pleura for the lung to expand** during deep inspiration.
- The **costomediastinal recess:** it lies anteriorly behind sternum and costal cartilages between costal and mediastinal pleurae and related to cardiac notch on left side.
- The **costodiaphragmatic recess:** it lies inferiorly between costal and diaphragmatic pleurae and it extands vertically from 8th to 10th ribs along mid-axillary line.



Blood supply of Pleura:

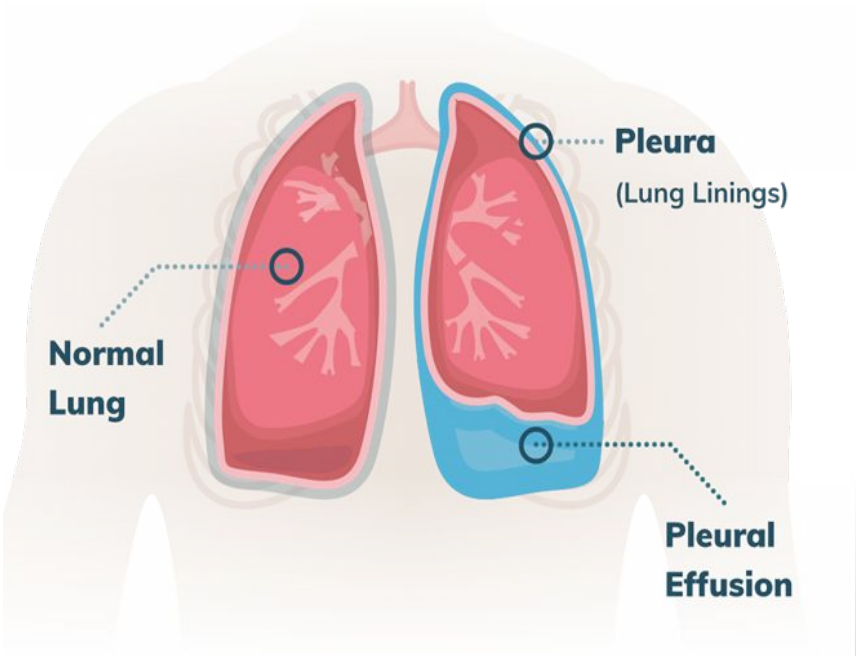
- Parietal pleura is supplied by intercostal, internal thoracic and musculophrenic arteries and veins mostly drain into azygous and internal thoracic veins.
- Pulmonary (visceral) pleura is supplied by bronchial arteries and veins drain into bronchial veins.

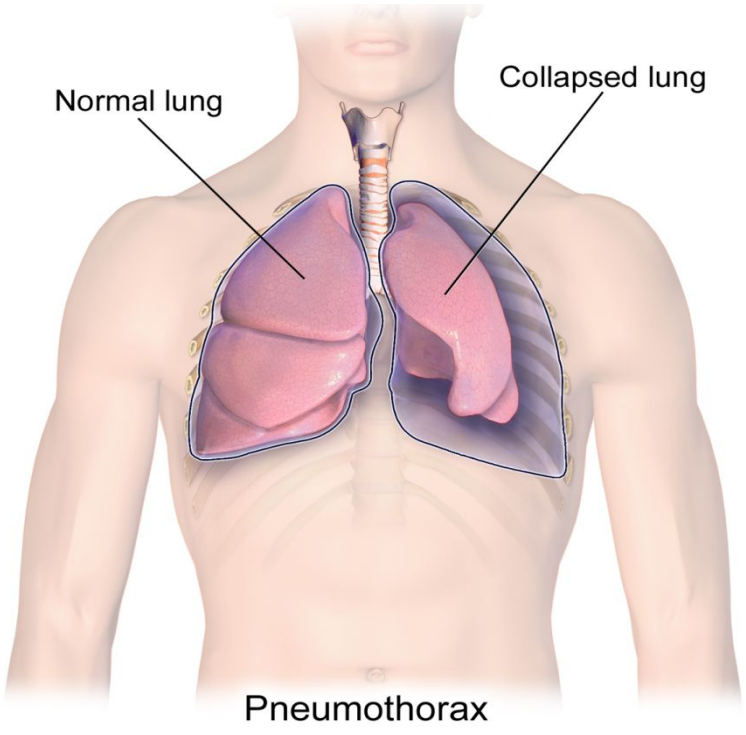


Differences between Visceral and Parietal Pleurae		
	Visceral	Parietal
Development position	Splanchnopleuric mesoderm, lines the surface of lung including fissures	Somatopleuric mesoderm, lines the thoracic wall, mediastinum and diaphragm
Nerve supply	Sympathetic nerves from T2-T5 ganglia, parasympathetic from vagus nerve	Thoracic nerves and phrenic nerves
Sensitivity	Insensitive to pain	Sensitive to pain
Blood supply	Bronchial vessels	Intercostal and pericardiophrenic vessels
Lymph drainage	Tracheobronchial lymph nodes	Intercostal lymph nodes

Applied Anatomy:

- **Pleural effusion:** Accumulation of excess of fluid between layers of pleura and outside the lungs.
- **Pleurisy:** This is the inflammation of pleura which may be dry or accompanied by pleural effusion.
- **Pneumothorax:** Presence of air in the pleural cavity.
- **Hydropneumothorax:** Presence of both fluid and air in the pleural cavity.
- **Hemothorax:** Accumulation of blood within the pleural cavity.
- **Empyema:** Presence of pus in the pleural cavity.






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